

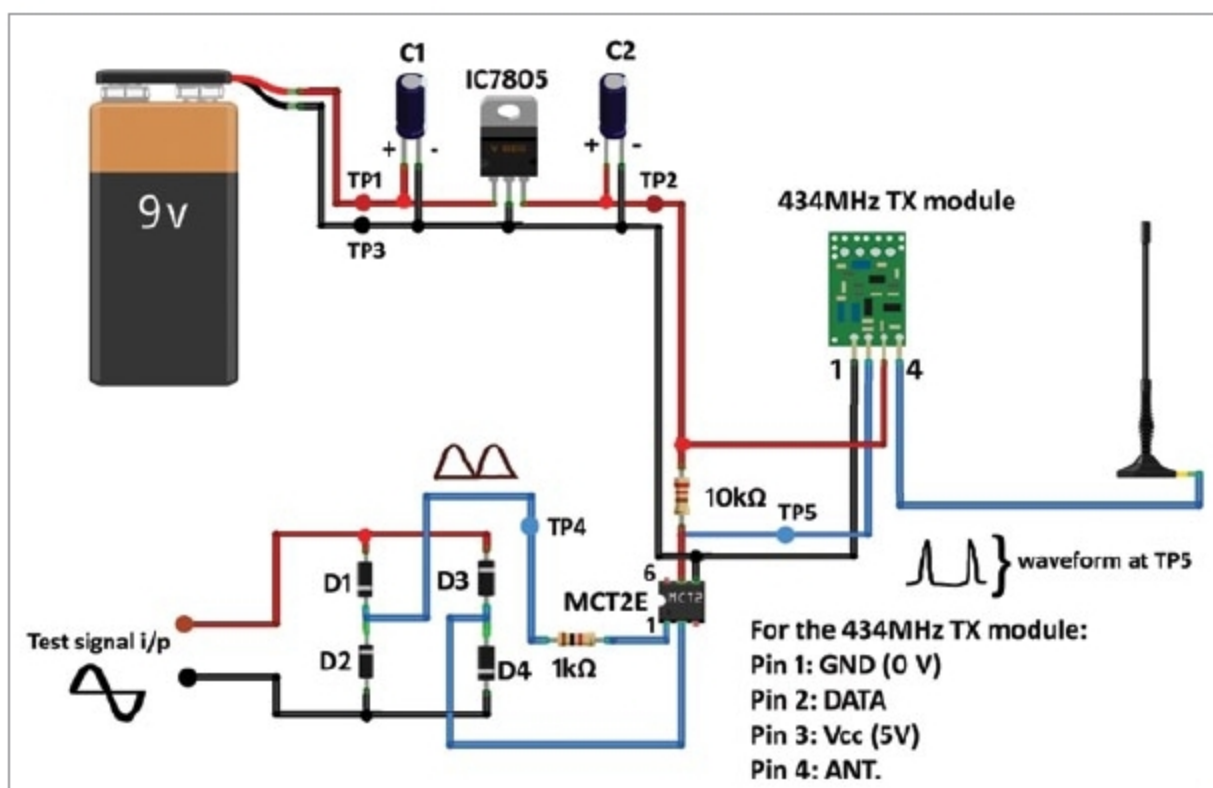


Fig. 5: Circuit diagram of the transmitter side

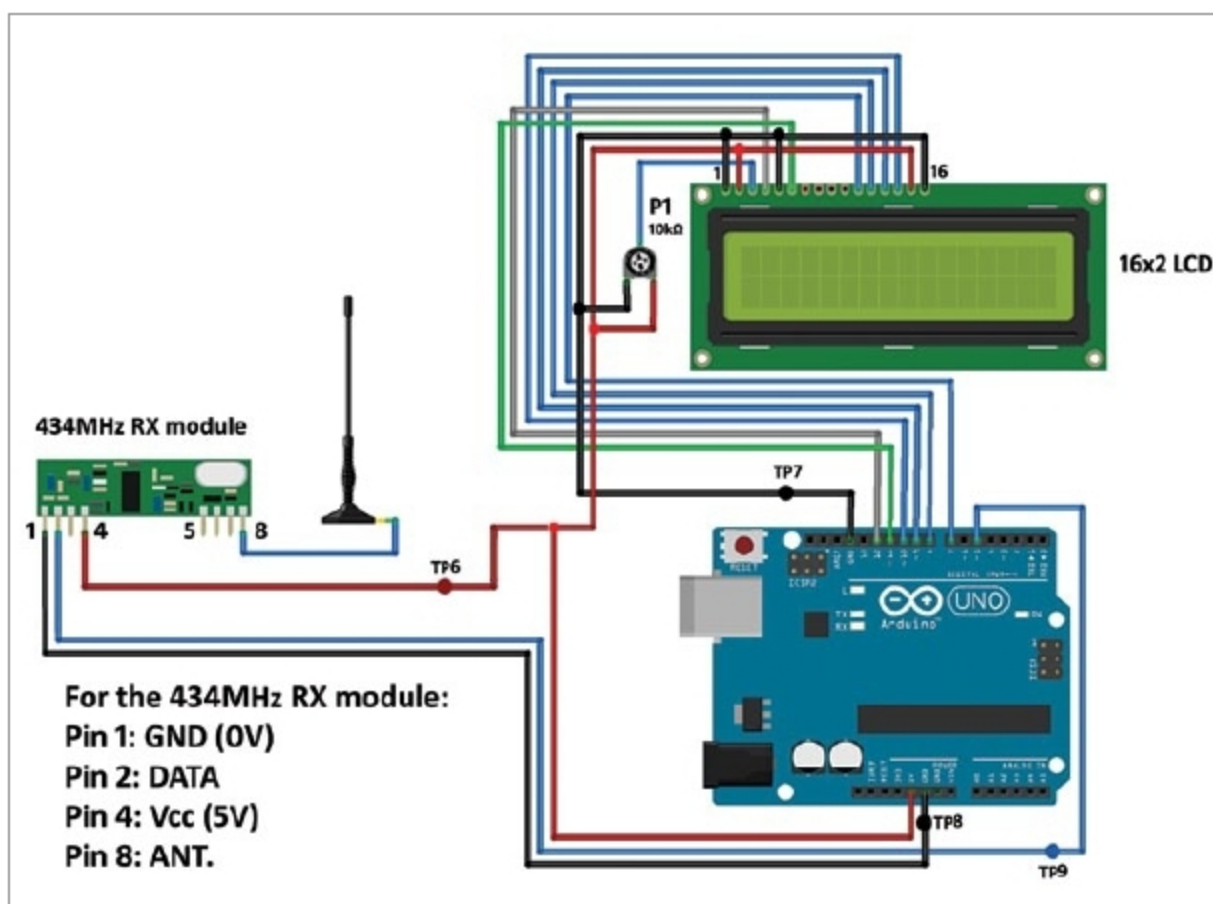


Fig. 6: Circuit diagram of the receiver side

On receiver side, 434MHz RX module demodulates the received signal. DATA pin of 434MHz RX is connected to digital I/O pin 5 of Arduino Uno, which calculates frequency of the test signal and displays it on the LCD and serial monitor of Arduino IDE. Test signal frequencies 1576Hz and 2399Hz displayed on the serial monitor are shown in Fig. 7.

This project can be used for experimenting, learning, testing and troubleshooting audio equipment in the audible range between 50Hz and 3kHz.

Components used in the project are listed in Table 1.

Software

Arduino IDE 1.6.5 is used for programming Arduino Uno. Select the proper COM port and board from Tools menu in the IDE. Upload freq_meter_wl_1.

EFY Note

The source code of this project is available for free download at source.efymag.com

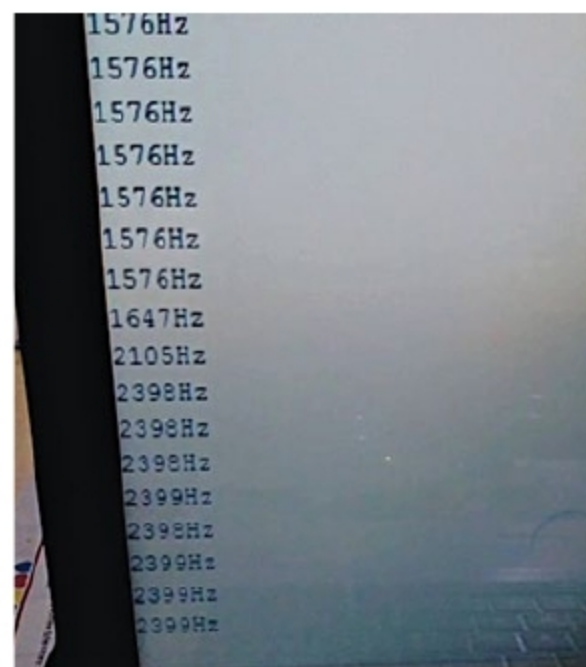


Fig. 7: Test signal frequencies (1576Hz and 2399Hz) displayed on the serial monitor

Expected Signals at Different Test Points

Table 2

Test point	Expected signal
TP1	9V
TP2	5V
TP3	GND (0V)
TP4	Pulsating DC signal
TP5	Train of signals
TP6	5V
TP7	GND (0V)
TP8	GND (0V)
TP9	Train of signals

ino source code to the board. From Tools menu of Arduino IDE, open the serial monitor and select the proper baud rate (9600 is used here) to display the frequency on the serial monitor.

For troubleshooting, test the signals at different test points marked in the circuit diagrams. Expected voltages at different test points are listed in Table 2. **EFY**



Saikat Patra is passionate about electronics and microcontroller-based embedded system applications



Shibendu Mahata is M.Tech (gold medallist) in instrumentation and electronics engineering from Jadavpur University. He is interested in designing microcontroller-based real-time embedded signal processing and process control systems